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SOLUTIONS

PHYSICS

91. When car overtakes motorcycle, both have travelled the same distance in the same time. Let the total distance travelled be S and the total time taken to overtake be t.

For motor cycle:

Maximum speed attained=36km/h

$= 10 m/s$

Since its acceleration = 1.0 m/s^2, the time t_1 taken by it to attain the maximum speed is given by $v = u + at_1$ ($\because u = 0$)

$10 = 0 + 1.0 \times t_1$; $t_1 = 10$ s

The distance covered by motorcycle in attaining the maximum speed is

$$S_1 = 0 + \frac{1}{2}at_1^2 = \frac{1}{2} \times 1.0 \times (10)^2 = 50m$$

The time during which the motorcycle moves with maximum speed is (t – 10) s.

The distance covered by the motorcycle during this time is

$S'_1 = 10 \times (t - 10) = (10t - 100)$ m

$\therefore$ Total distance travelled by motorcycle in time t is $S = S_1 + S'_1 = 50 + (10t - 100)$

$= (10t - 50)$ m .....(1)

For car :

Maximum speed attained = 54 km h^{-1}

$= 54 \times \frac{5}{18} = 15$ m/s

Since its acceleration = 0.5 m/s^2

The time taken by it to attain the maximum speed is given by

$15 = 0 + 0.5 \times t_2$ ($\because u = 0$)

or $t_2 = 30$ s

The distance covered by the car in attaining the maximum speed is

$$S_2 = 0 + \frac{1}{2}at_2^2 = \frac{1}{2} \times 0.5 \times (30)^2 = 225m$$

The time during which the car moves with maximum speed is (t – 30) s.

The distance covered by the car during this time is $S'_2 = 15 \times (t - 30) = (15t - 450)$ m

$\therefore$ Total distance travelled by car in time t is, $S = S_2 + S'_2 = 225 + (15t - 450)$

$= (15t - 225)$ m

From equations (1) and (2), we get

$10t - 50 = 15t - 225$ or $5t = 175$

or $t = 35$ s

92. $\frac{1}{\sqrt{\mu_0 \varepsilon_0}} = c$; $\frac{1}{\mu_0 \varepsilon_0} = c^2$

93. Mass, m = 4.237 g, Volume, V = 2.5 cm^3

$$\text{Density, } \rho = \frac{\text{Mass}}{\text{Volume}} = \frac{4.237g}{2.5cm^3}$$

$= 1.6948$ g cm^{-3}

As mass has 4 significant figures and volume has 2 significant figures, therefore, as per rule, density will have only two significant figures. Rounding off to two significant figures, we get

$\rho = 1.7$ g cm^{-3}

94. $$\frac{h}{102} = \frac{\sin^2 30^\circ}{\sin^2 60^\circ} = \frac{\left(\frac{1}{2}\right)^2}{\left(\frac{\sqrt{3}}{2}\right)^2}$$

or $h = 102 \times \frac{1/4}{3/4} = 34m$

95. $2kx \cos 30^0 = \left(\frac{4m_1m_2}{m_1 + m_2}\right)g$

96. $a_r = \frac{20 \times 2 - 20 \times 10 \times 0.15}{20} = 0.5 m/s^2$

$4 = \frac{1}{2} \times 0.5 \times t^2$; $t = 4$ s

Distance of truck $= \frac{1}{2} \times 2 \times 4 \times 4 = 16\text{m}$

97. NCERT

98. $(\text{mgSin}\theta - \mu\text{mgCos}\theta)\text{x} = \frac{1}{2}\text{Kx}^2$

$\mu = 0.125$

99. $\overrightarrow{r_{com}} = \frac{m_1\vec{r_1} + m_2\vec{r_2}}{m_1 + m_2}$

100. NCERT

101. $\alpha = \frac{\Delta L}{L_0(\Delta\theta)}$

$= \frac{0.19}{100(100-0)} = 1.9\times10^{-5}\ /°C$

$\gamma = 3\alpha = 5.7\times10^{-5}/°\text{C}$

102. In series rate of flow of heat is same

$\Rightarrow \frac{K_A A(\theta_1 - \theta)}{l} = \frac{K_B A(\theta - \theta_2)}{l}$

$\Rightarrow 3K_B(\theta_1 - \theta) = K_B(\theta - \theta_2)$

$\Rightarrow 3(\theta_1 - \theta) = (\theta - \theta_2)$

$\Rightarrow 3\theta_1 - 3\theta = \theta - \theta_2 \Rightarrow 4\theta_1 - 4\theta = \theta_1 - \theta_2$

$\Rightarrow 4(\theta_1 - \theta) = (\theta_1 - \theta_2)$

$\Rightarrow 4(\theta_1 - \theta) = 20 \Rightarrow (\theta_1 - \theta) = 5°C$

103. Q =dU + W

W = 0 since volume is constant

Q = du

$V_{rms} = 2V_{rms}$, $T^1 = 4T$

$dU = (n_1 + n_2)(C_v)_{mix}(T_2 - T_1)$

$= (n_1 C_{v_1} + n_2 C_{v2})(T_2 - T_1) = 3825\ \text{R}$

104. $1 \rightarrow 2$ = isochoric

$2 \rightarrow 3$ = isobaric

$3 \rightarrow 4$ = isochoric

$4 \rightarrow 1$ = isobaric

105. Equivalent spring constant of a wire is given by, $K = \frac{YA}{l}$

$K_{eq} = K_1 + K_2$

or $\frac{Y(2A)}{l} = \frac{Y_1 A}{l} + \frac{Y_2 A}{l}$

or $Y = \frac{Y_1 + Y_2}{2}$

106. Volumetric flow rate of liquid

$Q = au = a\sqrt{2gh}$

$= 2\times10^{-6} m^2 \times \sqrt{2\times10\times2}\ m/s$

$= 12.6\times10^{-6} m^3/s$

107. Radius of big drop $R' = n^{1/3}.r = 2^{1/3}.R$

$\frac{\text{Surfaceenergy before change}}{\text{Surfaceenergyafter change}}$

$= \frac{n4\pi R^2 T}{4\pi R'^2 T} = \frac{2.R^2}{2^{2/3}.R^2} = 2^{1/3}$

108. $X = A\sin\omega t$; $\frac{A}{2} = A\sin\omega t$; $\sin\omega t = \frac{1}{2}$

so phase $\omega t = 30°$ or $150°$

$\Delta\phi = 150° - 30° = 120°$ or $\frac{2\pi}{3}$

109. $v_0 = \frac{v_e}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$ km/s

110. From Kepler's third law,

$T^2 \propto r^3$ or $T \propto (r)^{3/2}$

$\therefore \quad \frac{T_2}{T_1} = \left(\frac{r_2}{r_1}\right)^{3/2}$

or $T_2 = T_1\left(\frac{r_2}{r_1}\right)^{3/2} = (365)\left(\frac{1}{2}\right)^{3/2}$

or $T_2 \approx 129$ days

111. $n \propto \sqrt{T} \Rightarrow \frac{n_1}{n_2} = \sqrt{\frac{T_1}{T_2}}$

$\Rightarrow \quad n_1\sqrt{T_2} = n_2\sqrt{T_1}$

$(n-10)\sqrt{160} = n\sqrt{129.6}$

112. $\vec{p} = p\cos\theta\hat{i} + p\sin\theta\hat{j}$

$\vec{E}_1 = E\hat{i}$

$\vec{T}_1 = \vec{p} \times \vec{E}_1$

$= \left(p\cos\theta\hat{i} + p\sin\theta\hat{j}\right) \times E\left(\hat{i}\right)$

$\tau\hat{k} = pE\sin\theta\left(-\hat{k}\right)$

$\vec{E}_2 = \sqrt{3}E_1\hat{j}$

$\vec{T}_2 = \left(p\cos\theta\hat{i} + p\sin\theta\hat{j}\right) \times \sqrt{3}E_1\hat{j}$

$-\tau\hat{k} = \sqrt{3}pE_1\cos\theta\hat{k}$

From (i) & (ii)

$pE\sin\theta = \sqrt{3}pE\cos\theta$

$\tan\theta = \sqrt{3}$

$\theta = 60^\circ$

113. For series combination

$C_s = \frac{C_1C_2}{C_1 + C_2} = \frac{2}{3}\mu F$

When connected in series the maximum charge that can flow through the combination equals the lower value of charge accommodated by the first capacitor i.e. 6000 μC

$\therefore Q_1 = 6000\ \mu C$ and $8000\ \mu C$

$V_s = \frac{Q_1}{C_s} = \frac{6000\mu C}{(2/3)\mu C} = 9kV$

114. Let r be the radius of each small drop and q coulomb be the charge on each.

Then potential

at the surface of each drop is $V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$

Volume of big drop = n × volume of small drop

$\frac{4}{3}\pi R^3 = n.\frac{4}{3}\pi r^3 \Rightarrow R = n^{1/3}\,r$ ..... (2)

From eq. (1), $q = V \times 4\pi\varepsilon_0 r$

Potential of big drop $V' = \frac{1}{4\pi\varepsilon_0}\frac{nq}{R}$

$\Rightarrow V' = \frac{n4\pi\varepsilon_0 \times V \times r}{4\pi\varepsilon_0 \times n^{1/3}r} = n^{2/3}V$

115. $15\times40 + 5\times100 + 5\times80 + 1000 = V \times I$

$600 + 500 + 400 + 1000 = 220I$

$I = \frac{2500}{220} = 11.36$

$I = 12\ A$

116. $\frac{R_1}{R_2} = \frac{I}{(100-I)}$

$\frac{R_2}{R_1} = \frac{(I-10)}{(110-I)}$

$(100-I)(110-I) = I(I-10)$

$11000 + R - 210l = R - 10l$

$\Rightarrow I = 55$ cm

$R_1 = R_2\left(\frac{55}{45}\right)$

$R_1 + R_2 = 1000\Omega$

$R_1 = 550\Omega$

117. $20 - 2i_1 - 2(i_1 + i_2) = 0 \Rightarrow 20 - 2i_1 = 10 - 4i_2$

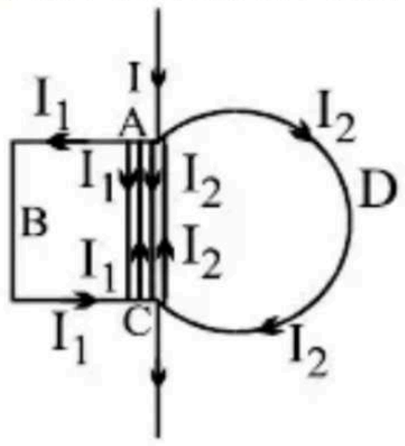
$10 - 4i_1 - 2(i_1 + i_2) = 0 \Rightarrow \frac{10 - 2i_1}{4} = i_2$

$\therefore 20 = 2i_1 + 2i_1 + 2i_2 \Rightarrow 20 = 4i_1 + 5 - i_1$

$\Rightarrow 3i_1 = 15 \therefore i_1 = 5$ and $i_2 = 0$

$i_1 + i_2 = 5\ A$

118. Introducing two equal and opposite current I_1 and also I_2 between A & C.



Force on ABCA closed loop zero

Force on ADCA closed loop zero

Force on extra I_1 & I_2

$F = (I_1 + I_2)\, lB = IlB$

119. This is equivalent to a current loop of radius R

$I = ne$

$$B = \frac{\mu_0 I}{2R} = \frac{\mu_0 ne}{2R}$$

120. For diamagnetic material, magnetic susceptibility is independent of temperature.

121. $$e = \frac{d\phi}{dt} = \frac{d[(B)\pi r^2]}{dt} = 2\pi rB\left(\frac{dr}{dt}\right)$$

$= (2\pi)(2 \times 10^{-2})(2 \times 10^{-3})\ (0.04) = 3.2\ \pi\mu V$

122. $V^2 = V_R^2 + (V_L - V_C)^2$

123. R Changed $\Rightarrow$ Z Changed $\Rightarrow$ I Changed

$V_L = I X_L$ (Changed)

$V_C = I X_C$ (Changed)

V_L and V_C are changed by same amount

Voltage across LC combination ($V_L - V_C$) remains same

124. Total average energy density of the electromagnetic $= \varepsilon_0 E^2_{rms}$

$= 4.58 \times 10^{-6}\ Jm^{-3}$

125. Focal length

$$f = \frac{-R}{2(\mu - 1)} = \frac{-10}{2(1.5 - 1)}$$

$f = -10$ cm

126. To produce dispersion without deviation

$(\mu_1 - 1)\, A_1 = (\mu_2 - 1) A_2$

$$\Rightarrow A_2 = \left(\frac{\mu_1 - 1}{\mu_2 - 1}\right) A_1 = \left(\frac{1.5 - 1}{1.75 - 1}\right)(15^\circ)$$

$= 10^\circ$

127. Conceptual

128. ncert

129. Einstein P.E. equation

Case I : $$eV = \frac{hc}{\lambda} - \frac{hc}{\lambda_0} \quad \ldots\ldots (1)$$

Case II :

$$e\frac{V}{4} = \frac{hc}{2\lambda} - \frac{hc}{\lambda_0} \Rightarrow eV = \frac{4hc}{2\lambda} - \frac{4hc}{\lambda_0} \quad \ldots\ldots (2)$$

Eq. (1) & eq. (2)

$$\frac{hc}{\lambda} - \frac{2hc}{\lambda} = -\frac{4hc}{\lambda_0} + \frac{hc}{\lambda_0}$$

$$\Rightarrow \frac{hc}{\lambda} = \frac{3hc}{\lambda_0} \Rightarrow \lambda_0 = 3\lambda$$

130. $m_e v_e = mv$

$$v = \frac{m_e v_e}{m} = \frac{9.1 \times 10^{-31} \times 3 \times 10^6}{10^{-6}}$$

$= 2.7 \times 10^{-18}$ m/s

131. Angular momentum is also called a moment of momentum.

For second orbit, $n = 2$; $L = \frac{2h}{2\pi} = \frac{h}{\pi}$

132. Here, $\Delta m = 0.3\%$ of 1 kg

$\frac{0.3}{100}\ kg = 3 \times 10^{-3}\ kg$

$E = (\Delta m)\, c^2 = 3 \times 10^{-3} \times (3 \times 10^8)^2$

$= 27 \times 10^{13}$ J

133. Height of potential barrier decreases when p-n junction is forward biased and it increases when junction is reverse biased.

134. Conceptual.

135. On the basis of the truth table, NOR gate

CHEMISTRY

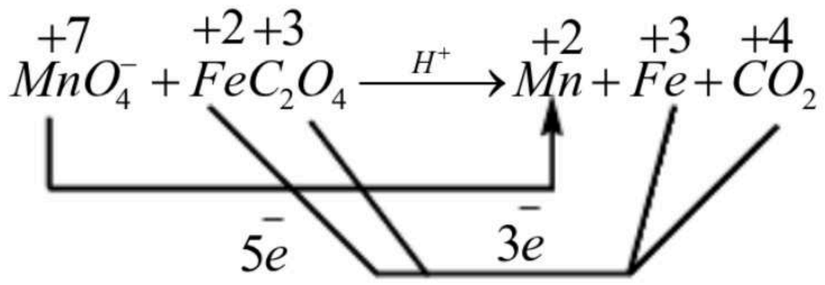
136. $\upsilon = \dfrac{2.19\times10^{8}}{n}.z$

137. Conceptual

138. Conceptual

139. C_6H_6 contains 12 planar atoms

140.



$\therefore 3MnO_4^- + 5FeC_2O_4$

$$\left.\begin{array}{l} 5FeC_2O_4 \rightarrow 3MnO_4^+ \\ 0.5FeC_2O_4 \rightarrow ? \end{array}\right\} 0.3$$

141. Zero between non zero digits are significant

142.

$18g\ water \rightarrow 40.79$

$80 \rightarrow ?$

$= \dfrac{80\times40.79}{18} \Rightarrow 181$

143. As 'P' is constant

$\Delta H = \Delta U + P\Delta U$

$= 30 + (2\times4)$

$= 38L.atm$

1L atm = 101.3 J

38L atm $\rightarrow$?

= 3849.4J

= 3.8494 KJ

144. $CO_2 + 2H_2O \rightleftharpoons H_3O^+ + HCO_3^-$

$$3.8\times10^{-7} = \frac{[H_3O^+][HCO_3^-]}{[CO_2]} \Rightarrow \frac{10^{-6}[HCO_3^-]}{[CO_2]}$$

$$\therefore \frac{[HCO_3^-]}{[CO_2]} = \frac{3.8\times10^{-7}}{10^{-6}} = 0.38$$

145. Conceptual

146. S for $AB = \sqrt{10^{-10}}$

S for $AB_2 = 3\sqrt{10^{-10}}$

S for $A_3B_2 = 5\sqrt{10^{-10}}$

S for $AB_3 = 4\sqrt{10^{-10}}$

147. Conceptual

148. $y_A = \dfrac{P_A^0\times A}{P}, y_B = \dfrac{P_B^0\times B}{P}$

$$\frac{y_A}{y_B} = \frac{P_A^0\times A}{P_B^0\times B} = \frac{4}{3} = \frac{1}{3}\times\frac{XA}{(1-XA)} \qquad XA = \frac{4}{5},\ XB = \frac{1}{5}$$

149. RLVP = iX_B

i = is highest for $AlCl_3$

150. $Al_2(SO_4)_3 = 2\wedge^0_{Al^{+3}} + 3\wedge^0_{SO_4^{-2}}$ = 2x + 3y

151. 4 mole of water = $(4\times18)=8GEW=8F$

$\therefore Q=i\times t$

$8\times96500=4\times t$

$t=2\times96500\text{sec}$

$=193000\text{sec}$

152. K value remain same as long as the temperature is same

153. Based on integrated rate equation

154. Conceptual

155. Value (V) Lysine(K) Tryptophan(W) are essential amino acids

156. Conceptual

157. CO_3^{2-} - Brisk effervescence with dil H_2SO_4

Sr^{+2} - Crimson red in flame test

158. (Geometrical isomerism), (Optical isomerism) and (Geometrical isomerism & Optical isomerism)

159. Li→Li

160. Most EN element is B among 13th group

161. $I^- < H_2O < NH_3 < CN^-$

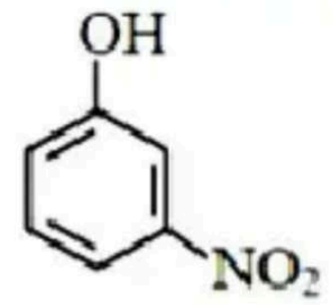
162. Conceptual

163. Wolf-Kishner reduction

164. free radical substitution

165. 4, 2, 2

166. i ii



167.

168. SN2 reaction at achiral carbon

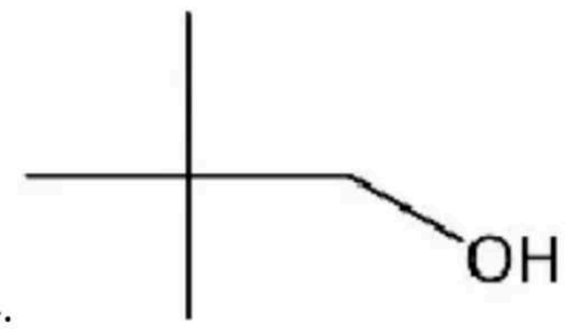
169. Hyperconjugation is a permanent effect

170. Dimond and Graphite

171. 40% formaldehyde solution

172. Pyrole

173. a-iii, b-iv, c-i, d-ii



174.

175. Stephens reaction

176. 6 and 7

177. All the metals except Sc forms 'MO' oxide

178. Trans isomer has higher melting point that cis isomer

179. A is not correct but R is correct

180. Due to the strong electron with drawing effect of the NO_2 group. The nucleophilicity of the $-NH_2$ is reduced and hence diazotization becomes difficult